[2024-2025 春夏] 离散数学理论基础 Quiz1

悔教 @cc98

2025 年 7 月 1 日

1. (35%) Determine whether the following statements are true or false.
   (5 points for a correct answer, 0 points for a blank answer, -2 points for an incorrect answer)

   a) If 'x is not occurring in A, then

   $$\exists x \big(P(x) \to A\big) = \forall x \big(P(x) \to A\big)$$

   ( )

   b) If A, B, and C are sets, then

   $$A - (B \cap C) = (A - B) \cup (A - C)$$

   ( )

   c) If n is an integer, then

   $$n = \lfloor n/2 \rfloor + \lceil n/2 \rceil$$

   ( )

   d) Suppose $P(x,y)$ is a predicate with universe $\{1,2,3,4\}$ and that

   $$P(1,3),\ P(2,1),\ P(2,4),\ P(3,2),\ P(3,4),\ P(4,1),\ P(4,4)$$

   are true; $P(x,y)$ is false for all other pairs. Then the statement

   $$\forall x \exists y \big((x \le y) \wedge P(x,y)\big)$$

   is true. ( )

   e)

   $$n^{0.01} \text{ is } O\Big((\log_{1.01} n)^{99999}\Big)$$

   ( )

   f) The set of positive real numbers less than 1 with decimal representations consisting only of 0s and 1s is countable. ( )

   g)

   $$2025^{2626} \equiv 1 \pmod{2027}$$

   ( )

2. (12%) Write a proposition equivalent to $p \oplus q$.

   a) Using only p, q, $\neg$, and the connective $\wedge$.

   b) Using only p, q, and the connective $|$ (where $|$ represents the NAND operation).

3. (9%) Find the full conjunctive normal form of $(p \oplus q) \vee r$.

4. (8%) Construct all functions from $A = \{1,2\}$ to $B = \{a,b\}$, and indicate which functions are bijections and which are surjections.

5. (9%) Given all positive integers relatively prime to 77 arranged in strictly increasing order, find the 600th term in this sequence.

6. (9%) Using the construction method from the Chinese Remainder Theorem proof, find all solutions to the system of congruences:

$$\begin{cases} x \equiv 1 \pmod 3, \\ x \equiv 2 \pmod 5, \\ x \equiv 3 \pmod 8. \end{cases}$$

7. (9%) Prove the distributive law for sets:

$$A_1 \cup (A_2 \cap \cdots \cap A_n) = (A_1 \cup A_2) \cap \cdots \cap (A_1 \cup A_n)$$

   for all $n > 2$.

8. (9%) Prove that every positive integer greater than 2 can be expressed as a sum of distinct Fibonacci numbers.